

Class Assignment 1: CS366

[Q1] [D]

[Q2] [A]

[Q3] [B][C]

[Q4] [D]

[Q5] [D]

[Q6] [A][B]

[Q7] [B]

[Q8] [A]

[Q9] [A][C]

[Q10] [A]

[Q11]

[a] **ANS:** 0.074

[b] **ANS:** Left

[c] **ANS:** $R(s) < -1.65$

[Q12]

[a] **ANS:** $U^{\pi_0}(1,1) = -0.85$ $U^{\pi_0}(1,2) = 0.75$

[b] **ANS:** $\pi_1(1,1) = \text{up}$ $\pi_1(1,2) = \text{right}$

[c] **ANS:** $U^{\pi_1}(1,1) = 0.66$ $U^{\pi_1}(1,2) = 0.918$

[d] **ANS:** Yes

[Q13]

a) $r \wedge \neg p$ **b)** $\neg p \wedge q \wedge r$ **c)** $r \rightarrow (q \leftrightarrow \neg p)$
d) $\neg q \wedge \neg p \wedge r$ **e)** $(q \rightarrow (\neg r \wedge \neg p)) \wedge \neg((\neg r \wedge \neg p) \rightarrow q)$
f) $(p \wedge r) \rightarrow \neg q$

[Q14]

$$R_6 : (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

$$R_7 : ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$

$$R_8 : (\neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})) .$$

$$R_9 : \neg(P_{1,2} \vee P_{2,1}) .$$

$$R_{10} : \neg P_{1,2} \wedge \neg P_{2,1} .$$

[Q15] Aima exercise 8.9

[a] ANS:

$In(Paris, France) \wedge In(Marseilles, France).$

(1) Correct.

[b] ANS:

$\exists c \text{ Country}(c) \wedge Border(c, Iraq) \wedge Border(c, Pakistan).$

(1) Correct.

[c] ANS:

$\forall c \text{ Country}(c) \wedge Border(c, Ecuador) \Rightarrow In(c, SouthAmerica).$

(1) Correct.

$\forall c \text{ Country}(c) \Rightarrow [Border(c, Ecuador) \Rightarrow In(c, SouthAmerica)].$

(1) Correct. Equivalent to (i).

[d] ANS:

$\neg[\exists c, d \text{ In}(c, SouthAmerica) \wedge In(d, Europe) \wedge Borders(c, d)].$

(1) Correct.

$\forall c, d [\text{In}(c, SouthAmerica) \wedge In(d, Europe)] \Rightarrow \neg Borders(c, d)].$

(1) Correct.

$\forall c \text{ In}(c, SouthAmerica) \Rightarrow \forall d \text{ In}(d, Europe) \Rightarrow \neg Borders(c, d).$

(1) Correct.

[e] ANS:

$\forall x, y \neg Country(x) \vee \neg Country(y) \vee \neg Borders(x, y) \vee$

$\neg(MapColor(x) = MapColor(y)).$

(1) Correct.

$\forall x, y (\text{Country}(x) \wedge \text{Country}(y) \wedge Borders(x, y) \wedge \neg(x = y)) \Rightarrow$

$\neg(MapColor(x) = MapColor(y)).$

(1) Correct. The inequality is unnecessary because no country borders itself.

[Q16]

(a)

$\neg P \rightarrow \neg\neg(Q \vee (R \wedge \neg S))$ Given

$\neg P \rightarrow (Q \vee (R \wedge \neg S))$ Double negation

$\neg\neg P \vee (Q \vee (R \wedge \neg S))$ Implication

$P \vee (Q \vee (R \wedge \neg S))$ Double negation

$(P \vee Q \vee R) \wedge (P \vee Q \vee \neg S)$ Distributivity

(b)

$(P \rightarrow (Q \vee (R \wedge S))) \wedge (R \vee (S \rightarrow Q))$ Given

$(P \rightarrow (Q \vee (R \wedge S))) \wedge (R \vee (\neg S \vee Q))$ Implication

$(\neg P \vee (Q \vee (R \wedge S))) \wedge (R \vee (\neg S \vee Q))$ Implication

$(\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee S) \wedge (R \vee \neg S \vee Q)$ Distributivity.

[Q17]

[a] ANS: $((R \wedge E) \Rightarrow C) \wedge (C \Rightarrow R) \wedge (C \Rightarrow E)$

[b] ANS: $(\neg R \vee \neg E \vee C) \wedge (\neg R \vee \neg C \vee E)$